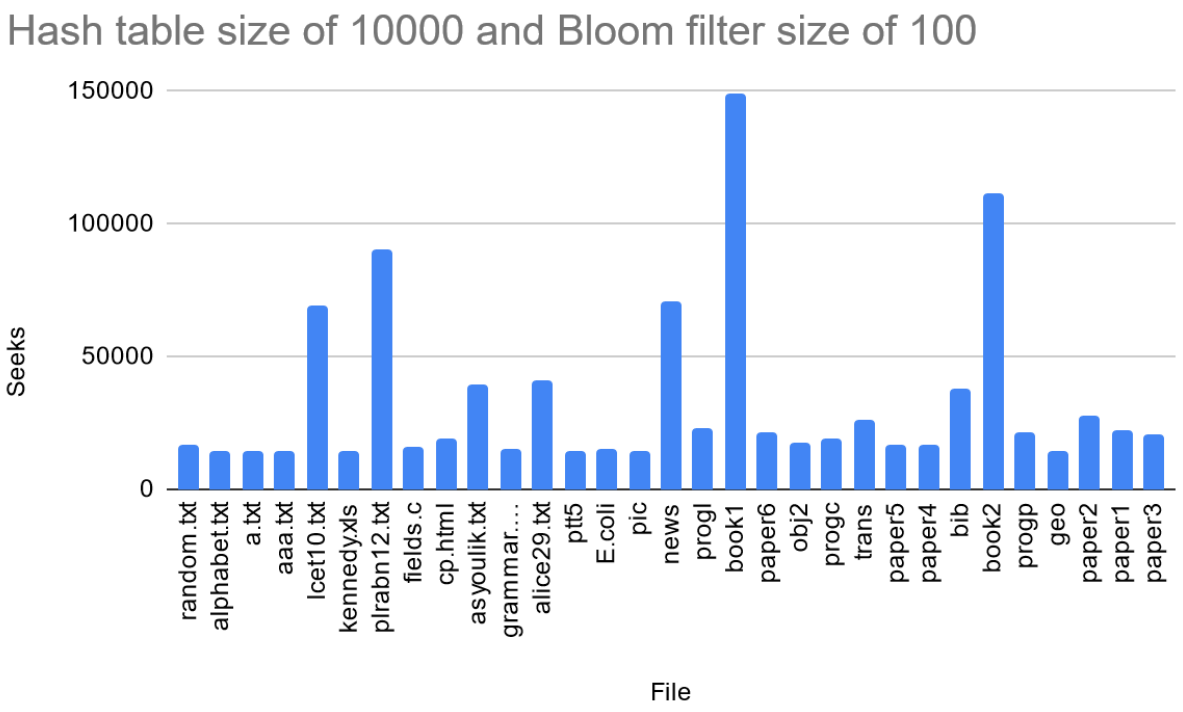
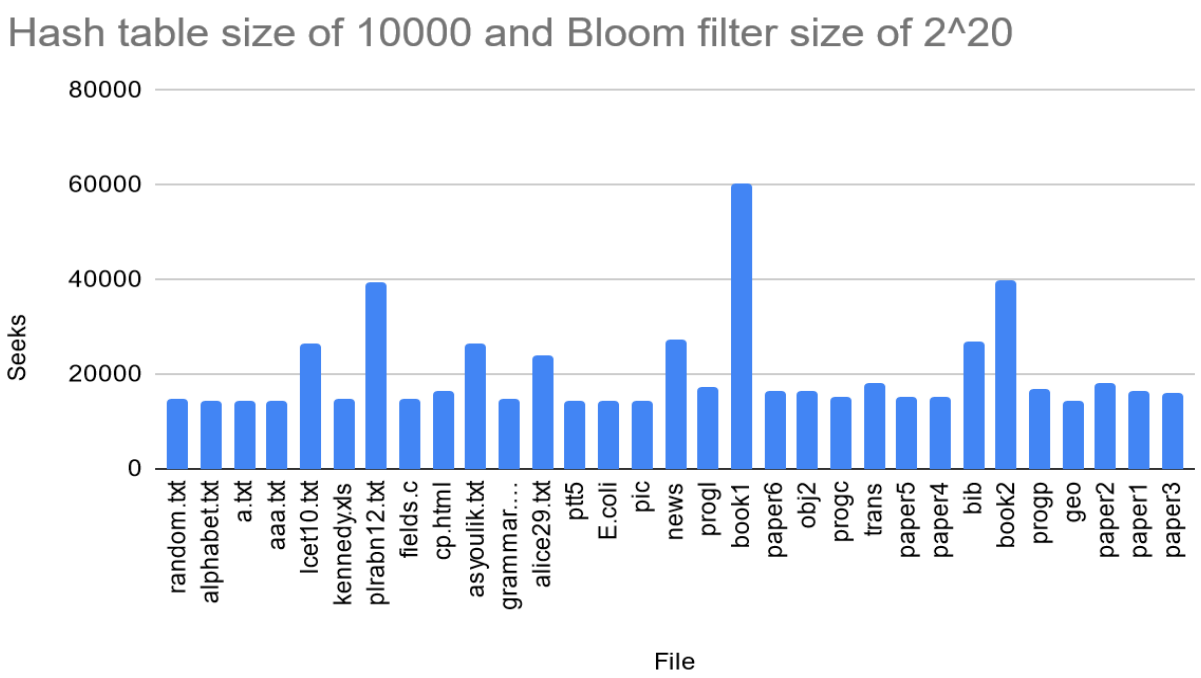
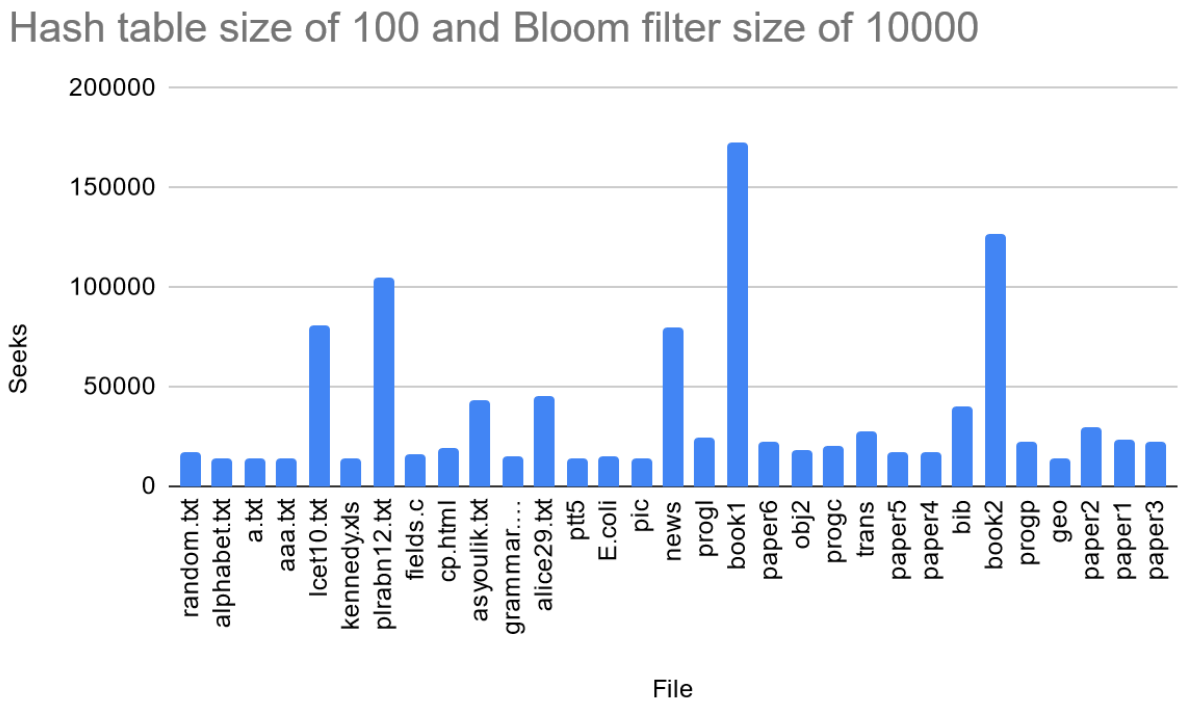
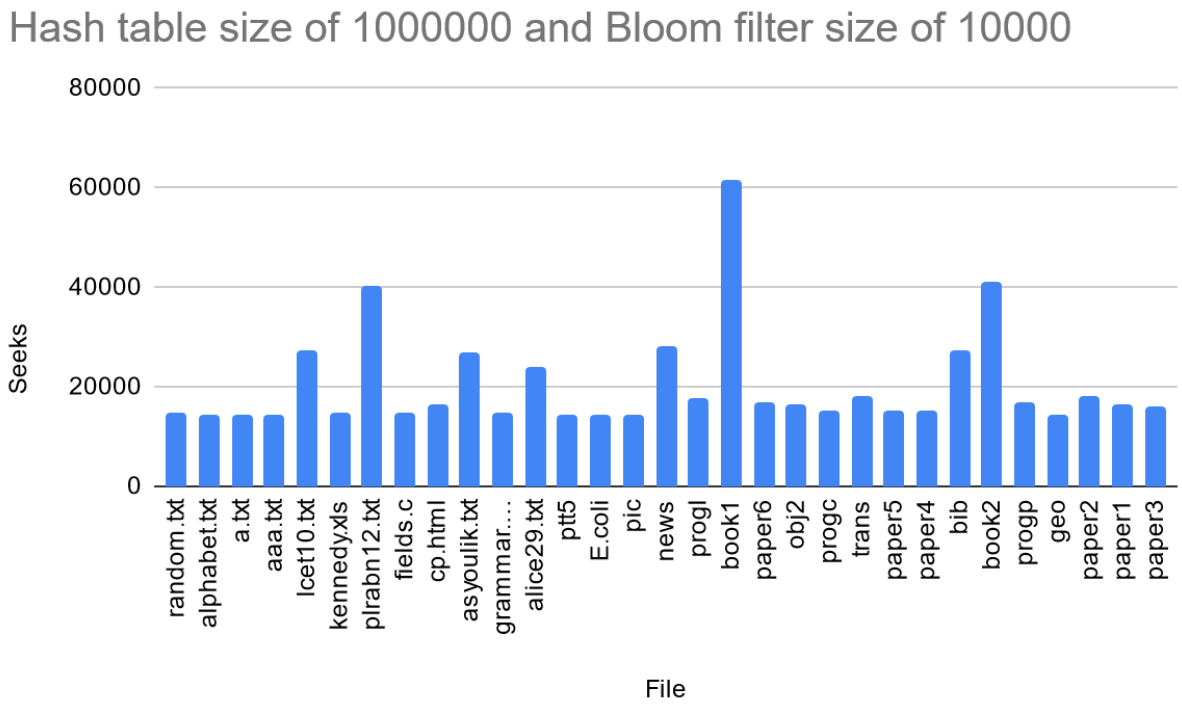
Darren Lee
CSE13S

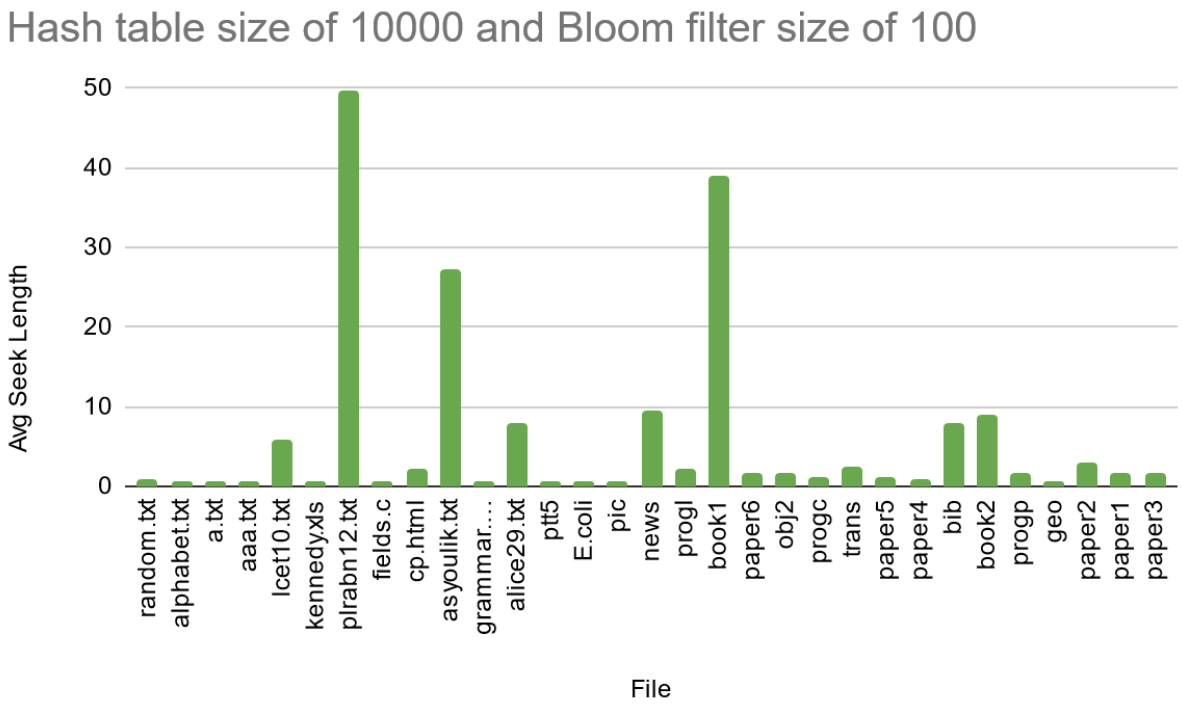
Assignment 7: The Great Firewall of Santa Cruz
Bloom Filters, Linked Lists and Hash Tables
Writeup

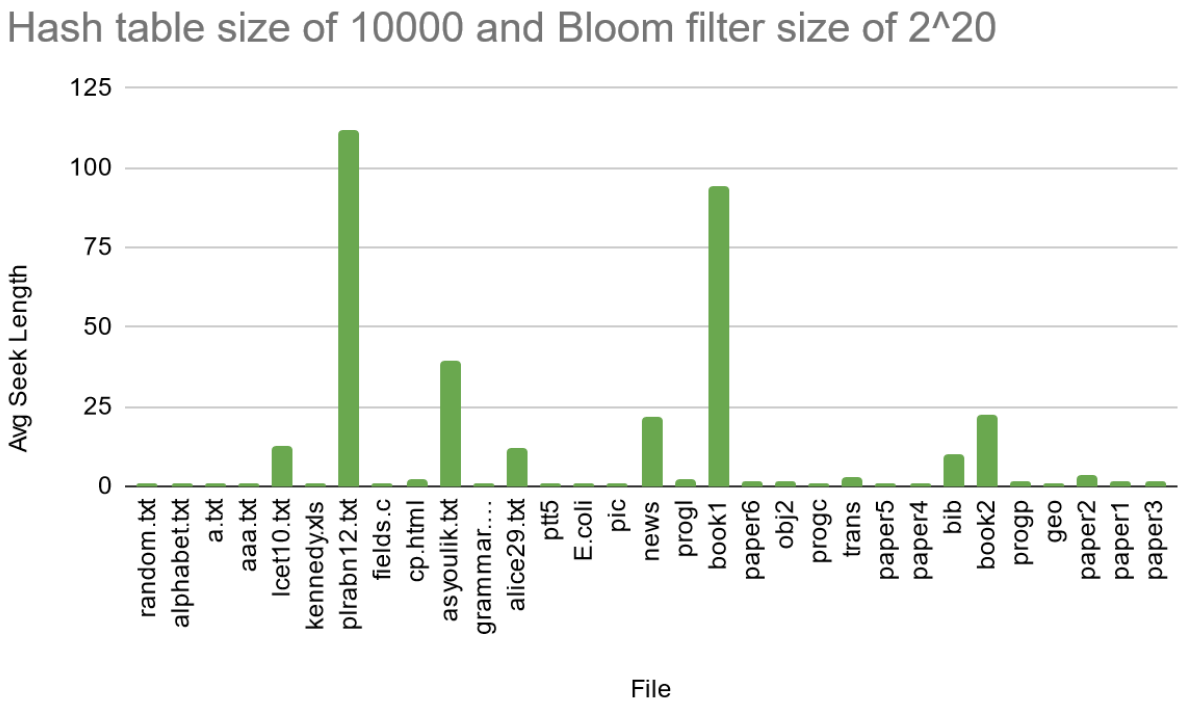


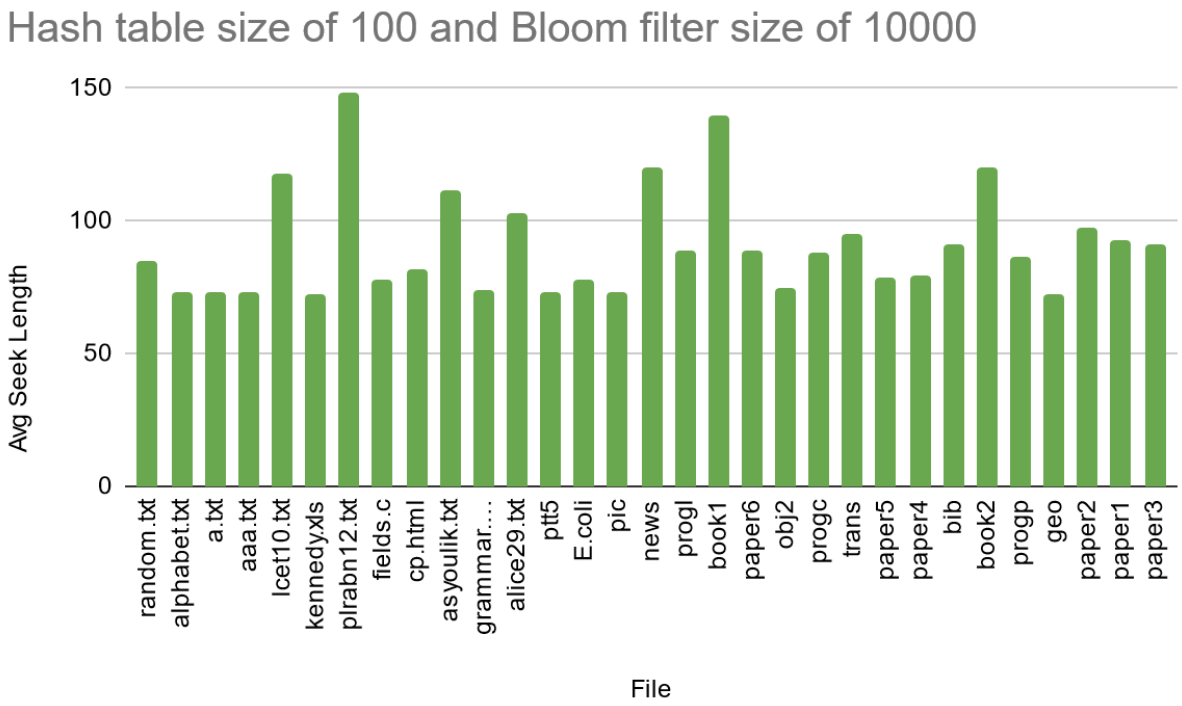


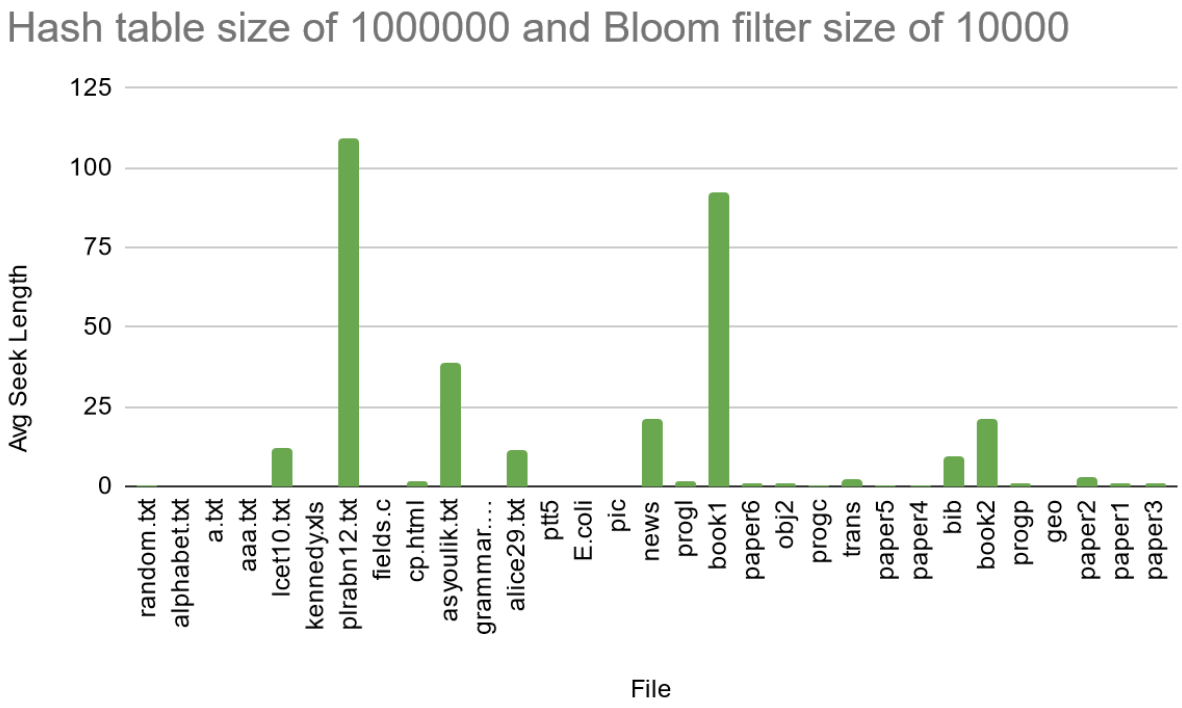

Hash table size of 100 and Bloom filter size of 10000
200000
150000
100000
50000
0
Seeks
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File


Hash table size of 1000000 and Bloom filter size of 10000
80000
60000
40000
20000
0
Seeks
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File


Hash table size of 10000 and Bloom filter size of 100
Avg Seek Length
50
40
30
20
10
0
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File


Hash table size of 10000 and Bloom filter size of 2^20
Avg Seek Length
125
100
75
50
25
0
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File


Hash table size of 100 and Bloom filter size of 10000
Avg Seek Length
150
100
50
0
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File

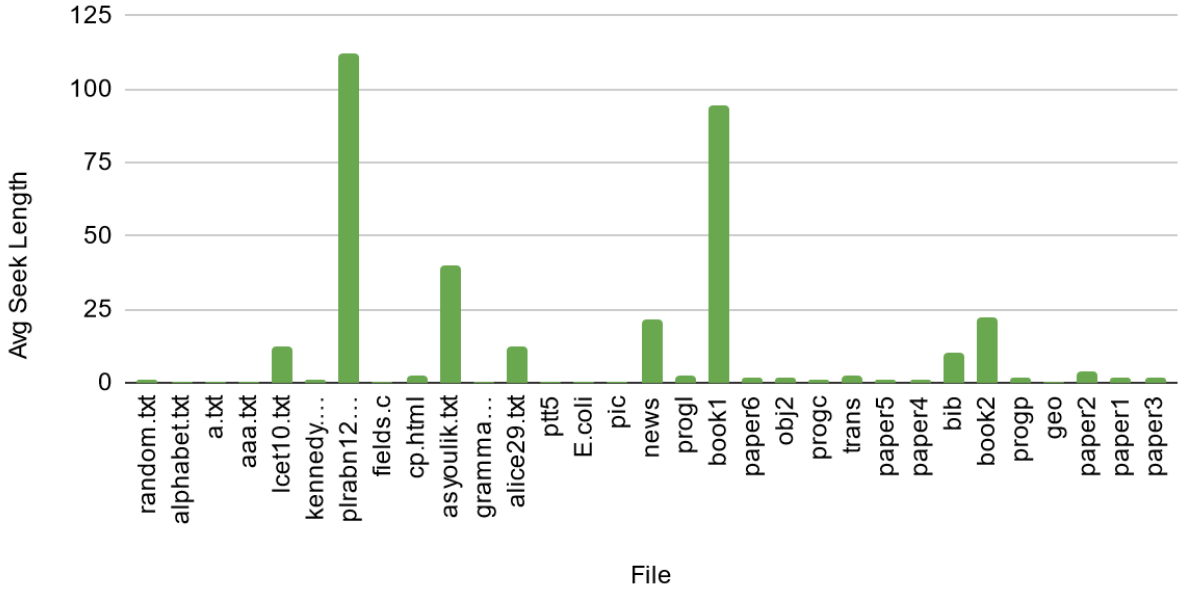

Hash table size of 1000000 and Bloom filter size of 10000
Avg Seek Length
125
100
75
50
25
0
random.txt
alphabet.txt
a.txt
aaa.txt
lcet10.txt
kennedy.xls
plrabn12.txt
fields.c
cp.html
asyoulik.txt
grammar....
alice29.txt
ptt5
E.coli
pic
news
progl
book1
paper6
obj2
progc
trans
paper5
paper4
bib
book2
progp
geo
paper2
paper1
paper3
File

Analysis

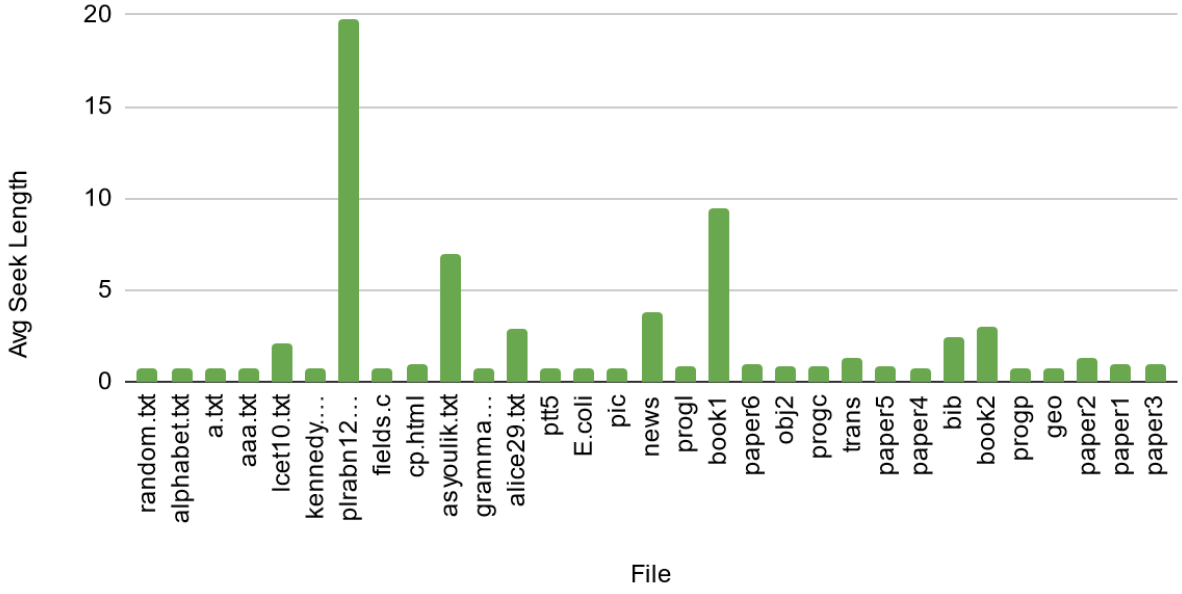
As the size of the Bloom filter increases when the hash table size stays the same, the number of seeks or lookups decreases. This is because a larger bloom filter has fewer false positives since there are more bits and a lower chance of the same bits being set by different hash functions. Since there are fewer false positives, the program does not need to perform a lookup of the hash table for the oldspeak words as often because bf_probe() does not falsely return true as often. The average seek length increases when the Bloom filter size increases but the hash table size stays the same. If the hash table size does not change, the linked lists will not get longer and the number of links will not change. The cause of the increase in average seek length is the decrease in the number of seeks performed. Since seek length is links/seeks, fewer seeks result in a greater average seek length. There are fewer seeks performed because there are fewer false positives when probing the Bloom filter.

As the size of the hash table increases, while the Bloom filter size stays the same, the number of seeks greatly decreases. This is due to the fact that the hash table can have a greater number of linked lists and whereas a small hash table will fill its hash table load and have to perform more lookups on the linked lists because every time a node is inserted, a linked list lookup is also performed to make sure the node is not already in the list. With fewer linked lists in the hash table, more nodes are inserted into each list and more lookups and seeks are performed. The average seek length also decreases with increased hash table size when the Bloom filter size remains unchanged because each linked list does not have to be as long when there are more linked lists total in the hash table. Shorter linked lists lead to fewer links traversed, and since seek length is links/seeks, fewer links allow for shorter average seek length.

Hash table size of 100000 and Bloom filter size of 2^20 with mtf disabled



Hash table size of 100000 and Bloom filter size of 2^20 with mtf enabled



Analysis

The move-to-front rule does not affect the number of seeks since the same lookups are required but greatly decreases the average seek length. This means that the number of links must be decreasing when the move-to-front rule is followed. The reason for this is that once a node is looked up, it does not require nearly as many links to be followed to be found again since it will be close to the beginning of the linked list. Commonly repeated words are quicker to find because they are near the front of the linked list. This means over the course of reading stdin, there are going to be fewer links overall when words show up repeatedly. If the words are extremely varied, then the move-to-front rule will not help much in decreasing the number of links because they will not be searched up again in the hash table. However, when there are words that are very common in a file and show up time after time, then move-to-front is especially useful for decreasing the number of links and therefore decreasing average seek length.